StatComp Project 2: Scottish weather

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Seasonal variability

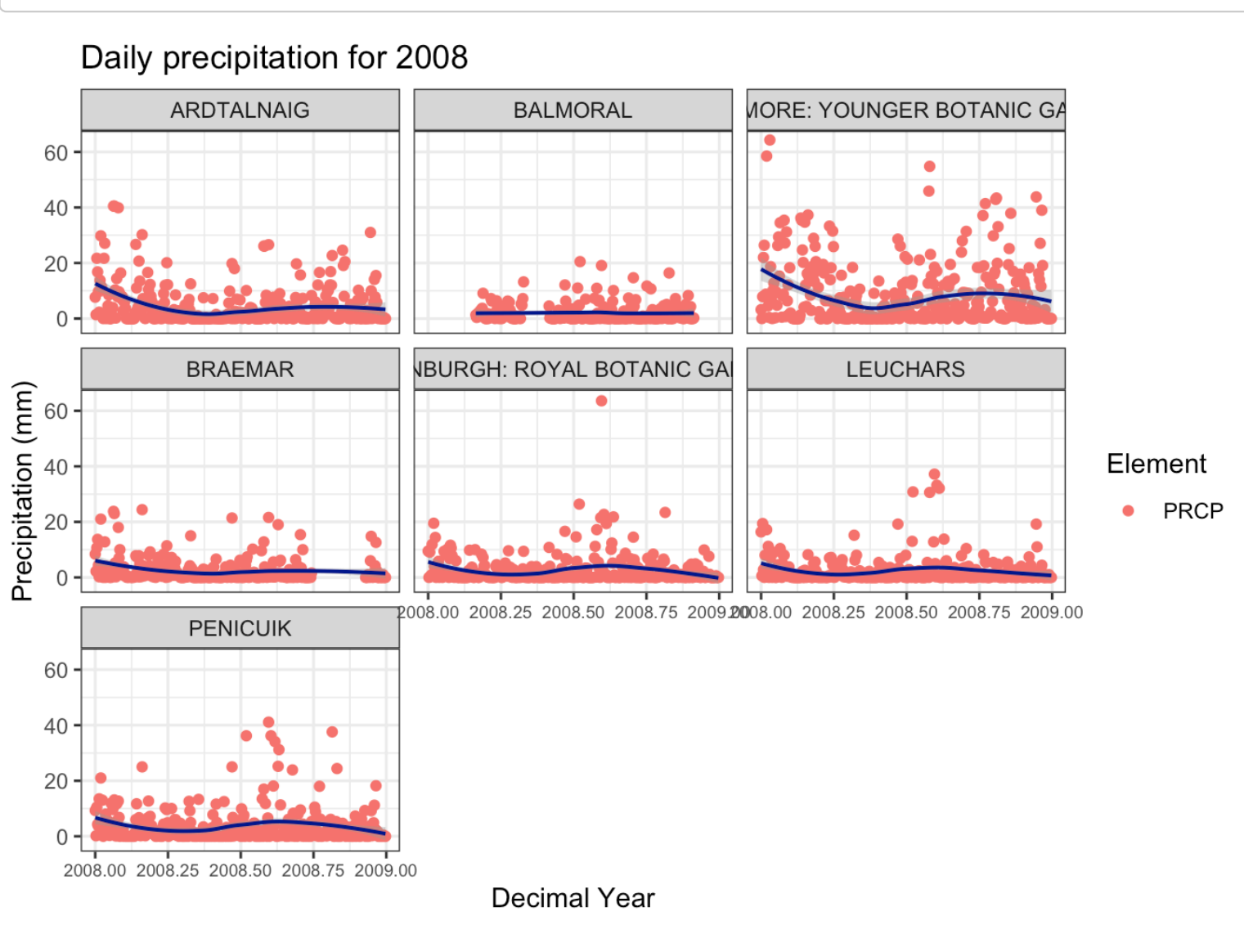
Is precipitation seasonally varying?

We aim to investigate the variability of Scottish perception and temperature data across a span of approximately sixty years by first loading the data **ghcnd_stations** and **ghcnd_values**, and then combine them with the `left_jiont` function into a new data frame called `ghcnd`. To facilitate analysis, a single year was selected for examination, specifically the year 2008. We filter the data on `Year == 2008` and create two plots to displaying the daily precipitation and temperature behaviors respectively.

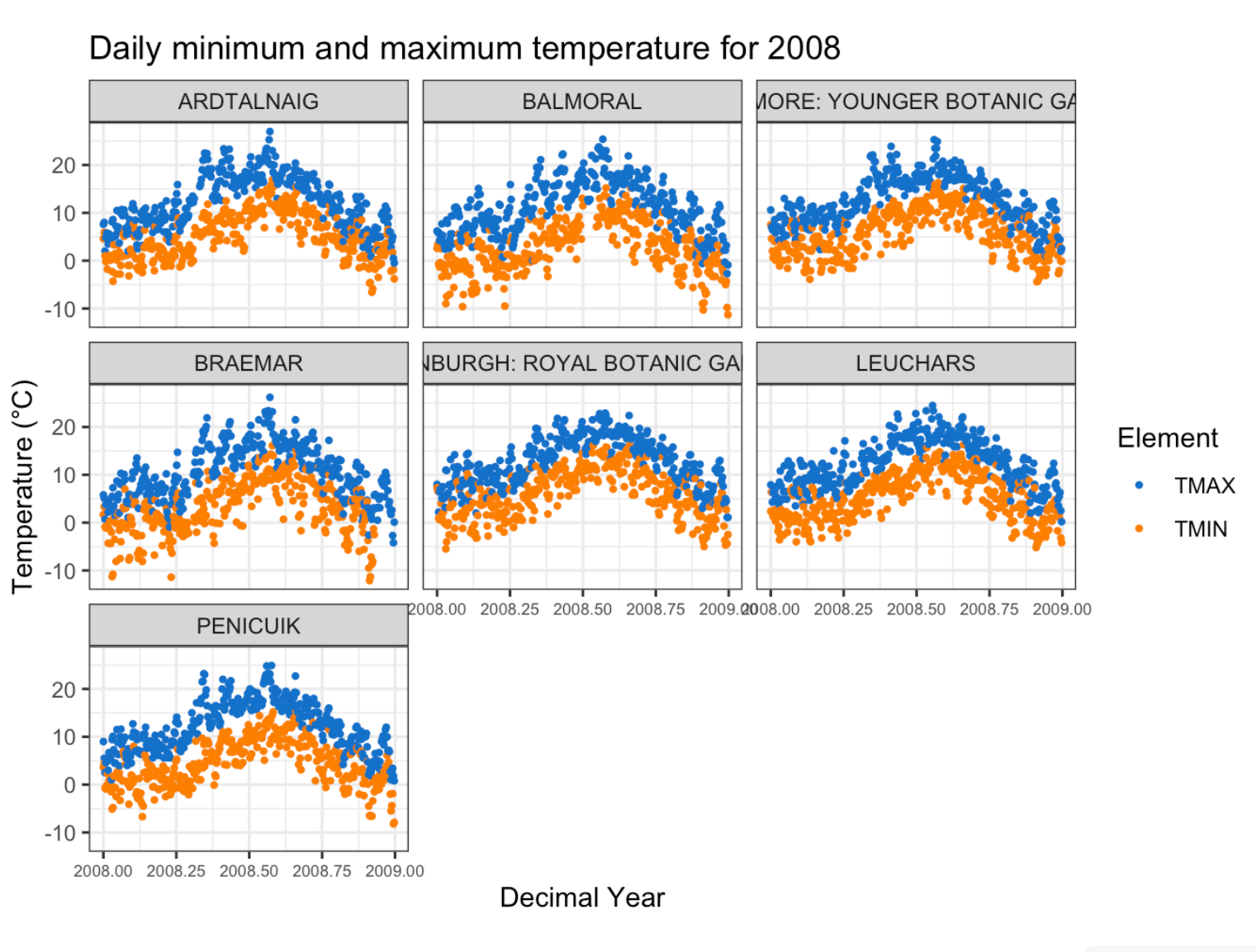
Generally, we first focus on the analysis on precipitation data by filtering on `Element == "PRCP"`. The result is shown on scatter plot where each dot represent the precipitation data for the specific day and there is a smooth line in darkblue shows the monthly average precipitation across the whole year. To be more specific, the plot shows the result for each station by applying `facet_wrap` function.

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Here we set April to September as summer and other month to be winter. The trend on these plots indicate that several station have more volatility, for instance Ardtalnaig and Benmore: Younger Bontanic Garden, whereas the others are more smooth. To be more specifically, Ardtalnaig and Benmore: Younger Bontanic Garden station exhibit higher amounts of precipitation during winter months and lower amounts during summer months, whereas the opposite is true for the other stations, except for Braemar, where the precipitation is consistent throughout the year.



Similarly to the previous step for precipitation, we select the temperature data by filtering on `Element %in% c("TMAX", "TMIN")`. It is evident that there is a seasonal effect with the temperature being always higher in summer and lower in winter. this trend was observed in all stations which exactly coincide with the seasonal effect above.

To further analyze the precipitation, there is Monte Carlo permutation test for the hypotheses that we want to test:

$$
\begin{aligned}
H_0 &: \text{The rainfall distribution is the same in winter as in summer} \\
H_1 &: \text{The winter and summer distributions have different expected values}
\end{aligned}
$$

We define the test statistics as the absoluate value of the difference between average precipitation in winter and summer for each station respectively as $T = |\text{winter average} - \text{summer average}|$. For convinence, we add a **Summer** column to the data that is **TRUE** for summer month and **FALSE** for winter month by the `mutate` function. We first calculate the test statistics T for each station.

Name	ID	winter_average	summer_average	T
ARDTALNAIG	UKE00105884	4.679448	2.668042	2.0114061
BALMORAL	UKE00105875	2.601891	1.994370	0.6075213
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	8.353198	4.995383	3.3578143
BRAEMAR	UKE00105874	2.878589	2.089790	0.7887997
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	1.869338	1.844564	0.0247740
FASKALLY	UKE00105885	2.899627	2.130559	0.7690686
LEUCHARS	UKE00105886	1.931070	1.809706	0.1213640
PENICUIK	UKE00105887	2.751264	2.326199	0.4250654

We do a randomized permutation test, since we have no model for the distribution of the data, by defining a function `p_value_CI` which calculate the p-value, Monte Carlo standard deviation and the confidence interval for p-value. Then, we count the number of times when the observed value(T_{obs}) is greater than or equal to the test statistic(T), $T_{obs} \geq T$. Further, the Porject2Hint document provided the model for the count, X, and the p-value as

$$X \sim \text{Bin}(N, p), \quad \text{and} \quad p = \mathbb{P}(T_{obs} \geq T | H_0).$$

where N represents the number of permutations we take. Then we have $\mathbb{E}(X) = Np$, and $\text{Var}(X) = Np(1-p)$. To do the further evaluation about the value of p, we can estimate it with $\hat{p} = x/N$ and thus implied that the expected value and variance of $\hat{p}$ are

$$\mathbb{E}(\hat{p}) = p, \quad \text{and} \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{N}.$$

Therefore, computing the p-values for each station and determining their Monte Carlo standard deviation can be accomplished by calculating x/N and taking the square root of their variance, which is given by $\sqrt{\frac{p(1-p)}{N}}$. Moreover, with sufficiently large values of N, we can calculate the 95% confidence interval for the p-values by

$$\text{CI} = \hat{p} \pm z_{0.975} \sqrt{\frac{p(1-p)}{N}}$$

Despite that most of the observed values of count is zero, resulting in a p-value approaching zero due to a fixed $N > 0$, we can instead calculate a confidence interval for this case by the interval $[0, 1 - 0.0025^{1/N}]$. By applying the `p_value_CI` function, the resulting table shows the p-value, Monte Carlo standard deviation, and corresponding confidence interval for the p-value of each station.

Name	p_value	monte_carlo_sd	lower_CI	upper_CI
ARDTALNAIG	0.0000	0.0000000	0.0000000	0.0003688
BALMORAL	0.0000	0.0000000	0.0000000	0.0003688
BENMORE: YOUNGER BOTANIC GARDE	0.0000	0.0000000	0.0000000	0.0003688
BRAEMAR	0.0000	0.0000000	0.0000000	0.0003688
EDINBURGH: ROYAL BOTANIC GARDE	0.6652	0.0047192	0.6559505	0.6744495
FASKALLY	0.0000	0.0000000	0.0000000	0.0003688
LEUCHARS	0.0310	0.0017332	0.0276030	0.0343970
PENICUIK	0.0000	0.0000000	0.0000000	0.0003688

Here we set $N = 10000$ and rejected the null hypothesis for each station where the corresponding p-value is less than 0.05. The table shows that most stations have a p-value of zero, but EDINBURGH: ROYAL BOTANIC GARDE and LEUCHARS have positive p-values. Notably, the p-value for EDINBURGH: ROYAL BOTANIC GARDE is surprisingly large, which indicates that we failed to reject the null hypothesis of no seasonal difference in rainfall distribution. Conversely, for LEUCHARS, the p-value is positive but less than 0.05, allowing us to reject the null hypothesis. Therefore, except for EDINBURGH: ROYAL BOTANIC GARDE, we reject the null hypothesis for all stations, indicating that a seasonal difference in rainfall distribution exists at these seven stations.

Spatial weather prediction

Estimation and prediction

For the second part of the project, we begin by modifying our data frame with the code below. We calculate the square root of the monthly averaged precipitation value and store it as a new variable called **Value_sqrt_avg**. This generates a new data frame **ghcnd_spatial**.

```r
# Construct a version of the data set with a new variable, Value_sqrt_avg
# Subset the data to include only precipitation values
ghcnd_prcp<- ghcnd %>%
  filter(Element == "PRCP")
# Group by name, year, and month and compute the average value
ghcnd_spatial <- ghcnd_prcp %>%
  group_by(ID, Year, Month) %>%
  summarise(DecYear = mean(DecYear), Value_monthly = mean(Value), .groups = "drop") %>%
  mutate(Value_sqrt_avg = sqrt(Value_monthly))
```

Now there is a basic model for the square root of monthly average precipitation defined as

$$M_0 : \text{Value\_sqrt\_avg} \sim \text{Intercept} + \text{Longtitude} + \text{Latitude} + \text{Elevation} + \text{DecYear}$$

Then by adding covariates $\cos(2\pi kt)$ and $\sin(2\pi kt)$ of frequency $k = 1, 2, 3\ldots$ we can also model the seasonal variability, defining four models M_1, M_2, M_3, and M_4, where the predictor expression for model M_K adds

$$\sum_{k=1}^{K} [\gamma_{c,k} \cos(2\pi kt) + \gamma_{s,k} \sin(2\pi kt)]$$

to the M_0 predictor expression, for the model coefficient $\gamma_{c,k}$ and $\gamma_{s,k}$. Moreover, the time t here is defined to be the **DecYear**.

Now we aim to estimate the model parameters for all five models. We defined the function `fit_model` which takes the value of the corresponding data frame and the value of k, which respond to each model, and return the result of the fitted model. Further, with the help of `summary` function, we finally obtain the estimation of parameters for all models.

For model M_0 below, we found the estimate for longitude and latitude are both negative, which suggests that there is a slight decrease in precipitation as we move westwards and northwards. Moreover, the estimate for elevation and DecYear are both positive and very small, indicating that there is a slight increase in precipitation over the year with increasing elevation.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.4769477	1.3942775	1.776510	0.0757048
Longitude	-0.5464155	0.0107287	-50.930197	0.0000000
Latitude	-0.1359610	0.0201820	-6.736743	0.0000000
Elevation	0.0004022	0.0000769	5.230763	0.0000002
DecYear	0.0024109	0.0003968	6.076437	0.0000000

Then for the rest four models, additional coefficients for the cosine and sine functions are also included. These two coefficients represent the estimated change in the square root of monthly average precipitation for a 1 unit increase in time for a specific frequency of the $\cos$ and $\sin$ functions. To be more detailed, for model M_1, with frequency $k = 1$, the estimate for $\cos$ is about 0.183, which suggests that there is a increase in precipitation. Then for other models M_2, M_3 and M_4, we can generate the similar result as this with additional coefficients for higher frequency of the $\cos$ and $\sin$ functions. The table below shows the result for the estimation of parameters for the rest four models.

For M_1 model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.4899466	1.3190655	1.887660	0.0591249
Longitude	-0.5466656	0.0101494	-53.861761	0.0000000
Latitude	-0.1349611	0.0190924	-7.068842	0.0000000
Elevation	0.0004040	0.0000727	5.554431	0.0000000
DecYear	0.0023757	0.0003754	6.328596	0.0000000
cos(2 * pi * 1 * DecYear)	0.1828141	0.0089020	20.536227	0.0000000
sin(2 * pi * 1 * DecYear)	-0.1320307	0.0088699	-14.885257	0.0000000

For M_2 model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.4768245	1.3178582	1.8794316	0.0602390
Longitude	-0.5466570	0.0101400	-53.9107744	0.0000000
Latitude	-0.1349497	0.0190748	-7.0747777	0.0000000
Elevation	0.0004042	0.0000727	5.5617875	0.0000000
DecYear	0.0023820	0.0003750	6.3512162	0.0000000
cos(2 * pi * 1 * DecYear)	0.1829270	0.0088941	20.5672844	0.0000000
sin(2 * pi * 1 * DecYear)	-0.1320672	0.0088618	-14.9030438	0.0000000
cos(2 * pi * 2 * DecYear)	0.0308837	0.0088891	3.4743492	0.0005161
sin(2 * pi * 2 * DecYear)	0.0007686	0.0088659	0.0866895	0.9309215

For M_3 model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.4704214	1.3167785	1.8761100	0.0606937
Longitude	-0.5466809	0.0101317	-53.9575739	0.0000000
Latitude	-0.1349466	0.0190591	-7.0804395	0.0000000
Elevation	0.0004044	0.0000726	5.5697391	0.0000000
DecYear	0.0023850	0.0003747	6.3644191	0.0000000
cos(2 * pi * 1 * DecYear)	0.1827885	0.0088870	20.5681356	0.0000000
sin(2 * pi * 1 * DecYear)	-0.1320662	0.0088546	-14.9150677	0.0000000
cos(2 * pi * 2 * DecYear)	0.0307235	0.0088820	3.4590838	0.0005462
sin(2 * pi * 2 * DecYear)	0.0005964	0.0088587	0.0673209	0.9463287
cos(2 * pi * 3 * DecYear)	-0.0070723	0.0088684	-0.7974687	0.4252137
sin(2 * pi * 3 * DecYear)	0.0285383	0.0088721	3.2166209	0.0013047

For M_4 model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	2.4712330	1.3165058	1.8771151	0.0605559
Longitude	-0.5466744	0.0101296	-53.9681005	0.0000000
Latitude	-0.1349674	0.0190551	-7.0829934	0.0000000
Elevation	0.0004042	0.0000726	5.5682896	0.0000000
DecYear	0.0023852	0.0003747	6.3663294	0.0000000
cos(2 * pi * 1 * DecYear)	0.1828520	0.0088857	20.5783195	0.0000000
sin(2 * pi * 1 * DecYear)	-0.1320498	0.0088528	-14.9162192	0.0000000
cos(2 * pi * 2 * DecYear)	0.0307010	0.0088803	3.4572014	0.0005500
sin(2 * pi * 2 * DecYear)	0.0006781	0.0088570	0.0765625	0.9389744
cos(2 * pi * 3 * DecYear)	-0.0070451	0.0088668	-0.7945515	0.4269092
sin(2 * pi * 3 * DecYear)	0.0284900	0.0088705	3.2117859	0.0013268
cos(2 * pi * 4 * DecYear)	0.0135609	0.0088998	1.5237317	0.1276339
sin(2 * pi * 4 * DecYear)	0.0122274	0.0088379	1.3835220	0.1665617

Assessment: Station and season differences

To do the further investigation, we are looking for how well the model can predict precipitation at new location by constructing a stratified corss-validation. Here we define the `stratified_cv` function which takes two parameters, data frame and the value of k, which define one of the five models. The function will return a 96*5 dataframe shows the average SE and DS scores for each month by station after the corss validation process, including the Name, ID, Month, and its corresponding standard error scores(SE) and Dawid-Sebastiani scores(DS). Since there are 8 stations and 12 months, each of the five models will have exactly one result with $8 \times 12 = 96$ rows. Thus, there will be a total of $96 \times 5 = 480$ rows across all five models, which means a total of $2 \times 480 = 960$ number of values of scores(480 for SE score and 480 for DS scores).

For example, when choose $k = 1$, the result will return a table with 96 rows.

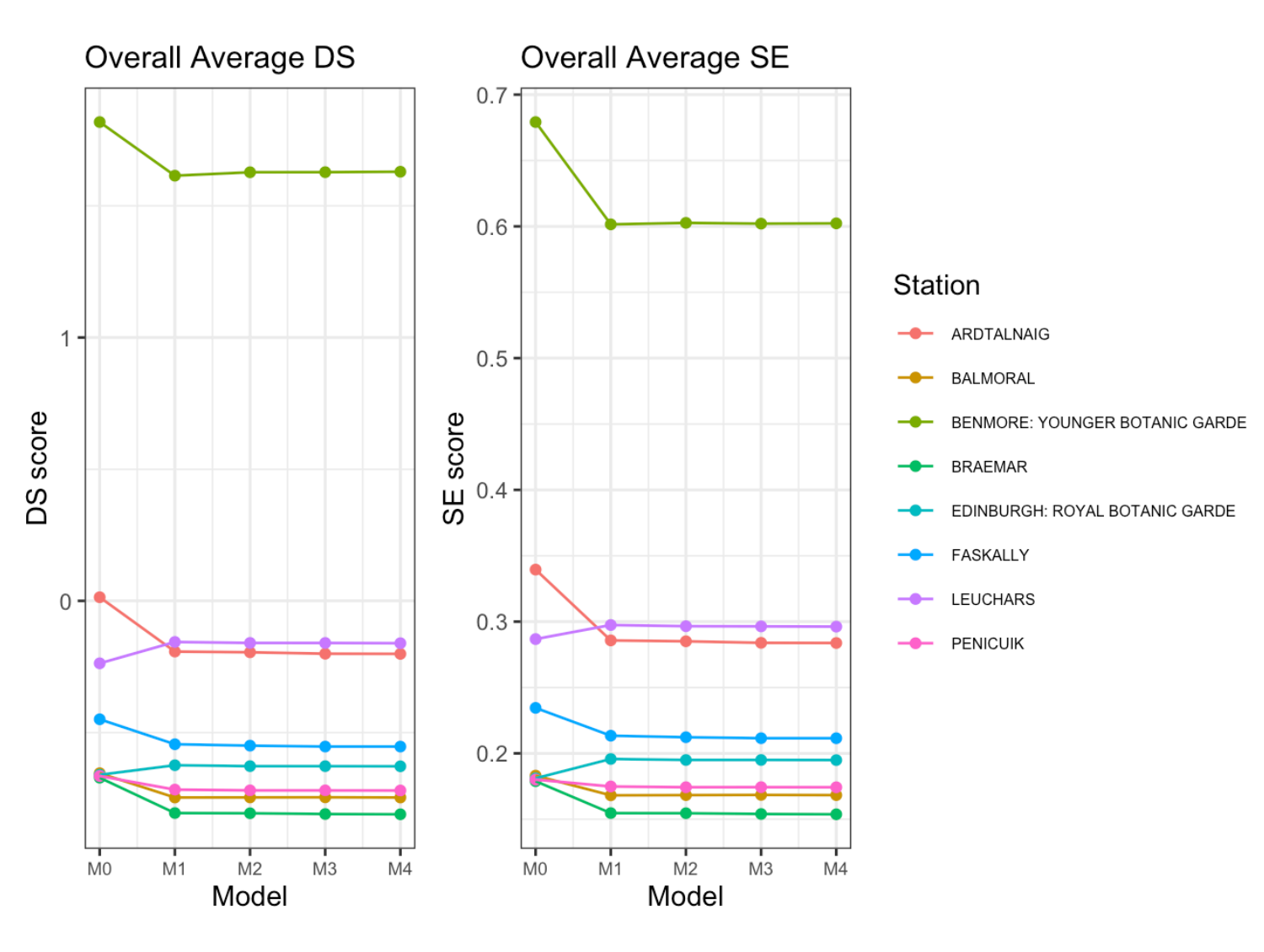
```r
stratified_cv(ghcnd_data, 1)
```

```
## # A tibble: 96 × 5
##    ID          Name    Month     se     ds
##    <chr>       <chr>   <int>  <dbl>  <dbl>
##  1 UKE00105874 BRAEMAR     1 0.185  -0.672
##  2 UKE00105874 BRAEMAR     2 0.159  -0.784
##  3 UKE00105874 BRAEMAR     3 0.0957 -1.07
##  4 UKE00105874 BRAEMAR     4 0.153  -0.811
##  5 UKE00105874 BRAEMAR     5 0.128  -0.924
##  6 UKE00105874 BRAEMAR     6 0.107  -1.02
##  7 UKE00105874 BRAEMAR     7 0.121  -0.958
##  8 UKE00105874 BRAEMAR     8 0.184  -0.675
##  9 UKE00105874 BRAEMAR     9 0.194  -0.629
## 10 UKE00105874 BRAEMAR    10 0.194  -0.629
## # … with 86 more rows
```

Moreover, we calculated the overall corss-validated average scores for each station with five models and hence get the result of a table of $5 \times 8 = 40$ rows.

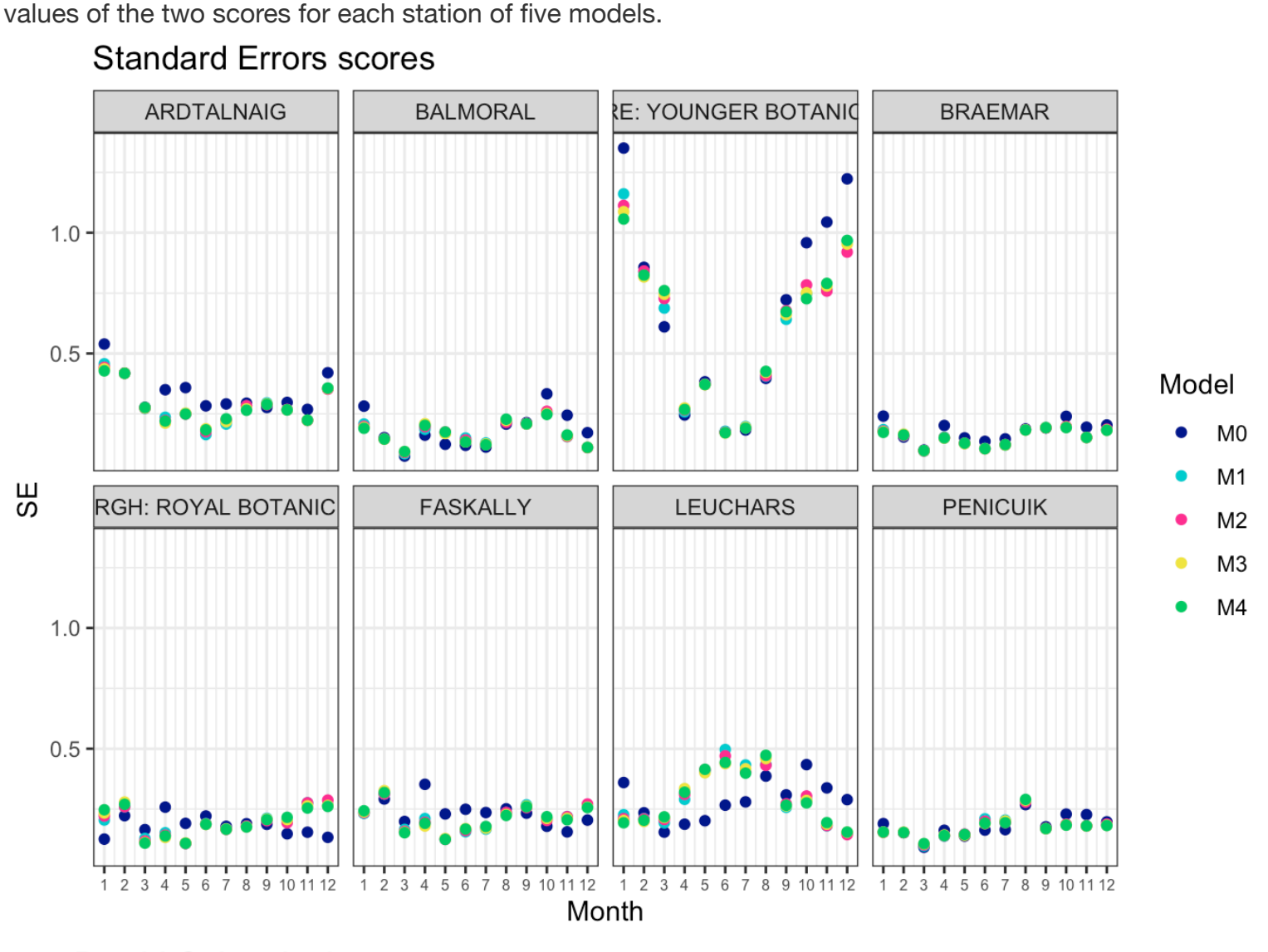
Name	ID	k	average_se	average_ds
ARDTALNAIG	UKE00105884	0	0.3394895	0.0137910
ARDTALNAIG	UKE00105884	1	0.2857313	-0.1925078
ARDTALNAIG	UKE00105884	2	0.2850669	-0.1952184
ARDTALNAIG	UKE00105884	3	0.2838975	-0.2006504
ARDTALNAIG	UKE00105884	4	0.2837743	-0.2012508
BALMORAL	UKE00105875	0	0.1830942	-0.6544201
BALMORAL	UKE00105875	1	0.1681612	-0.7466475
BALMORAL	UKE00105875	2	0.1683259	-0.7463112
BALMORAL	UKE00105875	3	0.1684852	-0.7459532
BALMORAL	UKE00105875	4	0.1683211	-0.7466557
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	0	0.6791761	1.8180702
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	1	0.6015445	1.6145002
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	2	0.6026575	1.6275220
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	3	0.6020913	1.6278410
BENMORE: YOUNGER BOTANIC GARDE	UKE00105930	4	0.6022941	1.6294953
BRAEMAR	UKE00105874	0	0.1788729	-0.6715428
BRAEMAR	UKE00105874	1	0.1546528	-0.8056632
BRAEMAR	UKE00105874	2	0.1545635	-0.8065269
BRAEMAR	UKE00105874	3	0.1539896	-0.8094017
BRAEMAR	UKE00105874	4	0.1537592	-0.8103960
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	0	0.1810967	-0.6604515
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	1	0.1957361	-0.6240794
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	2	0.1949797	-0.6276371
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	3	0.1949997	-0.6277090
EDINBURGH: ROYAL BOTANIC GARDE	UKE00105888	4	0.1948990	-0.6281629
FASKALLY	UKE00105885	0	0.2345134	-0.4496457
FASKALLY	UKE00105885	1	0.2134083	-0.5443998
FASKALLY	UKE00105885	2	0.2122443	-0.5497812
FASKALLY	UKE00105885	3	0.2115041	-0.5532186
FASKALLY	UKE00105885	4	0.2114887	-0.5532911
LEUCHARS	UKE00105886	0	0.2867391	-0.2372317
LEUCHARS	UKE00105886	1	0.2974184	-0.1561131
LEUCHARS	UKE00105886	2	0.2965283	-0.1598262
LEUCHARS	UKE00105886	3	0.2963989	-0.1599333
LEUCHARS	UKE00105886	4	0.2962063	-0.1608558
PENICUIK	UKE00105887	0	0.1800883	-0.6651038
PENICUIK	UKE00105887	1	0.1748678	-0.7167632
PENICUIK	UKE00105887	2	0.1742859	-0.7196454
PENICUIK	UKE00105887	3	0.1743311	-0.7197466
PENICUIK	UKE00105887	4	0.1742530	-0.7200933

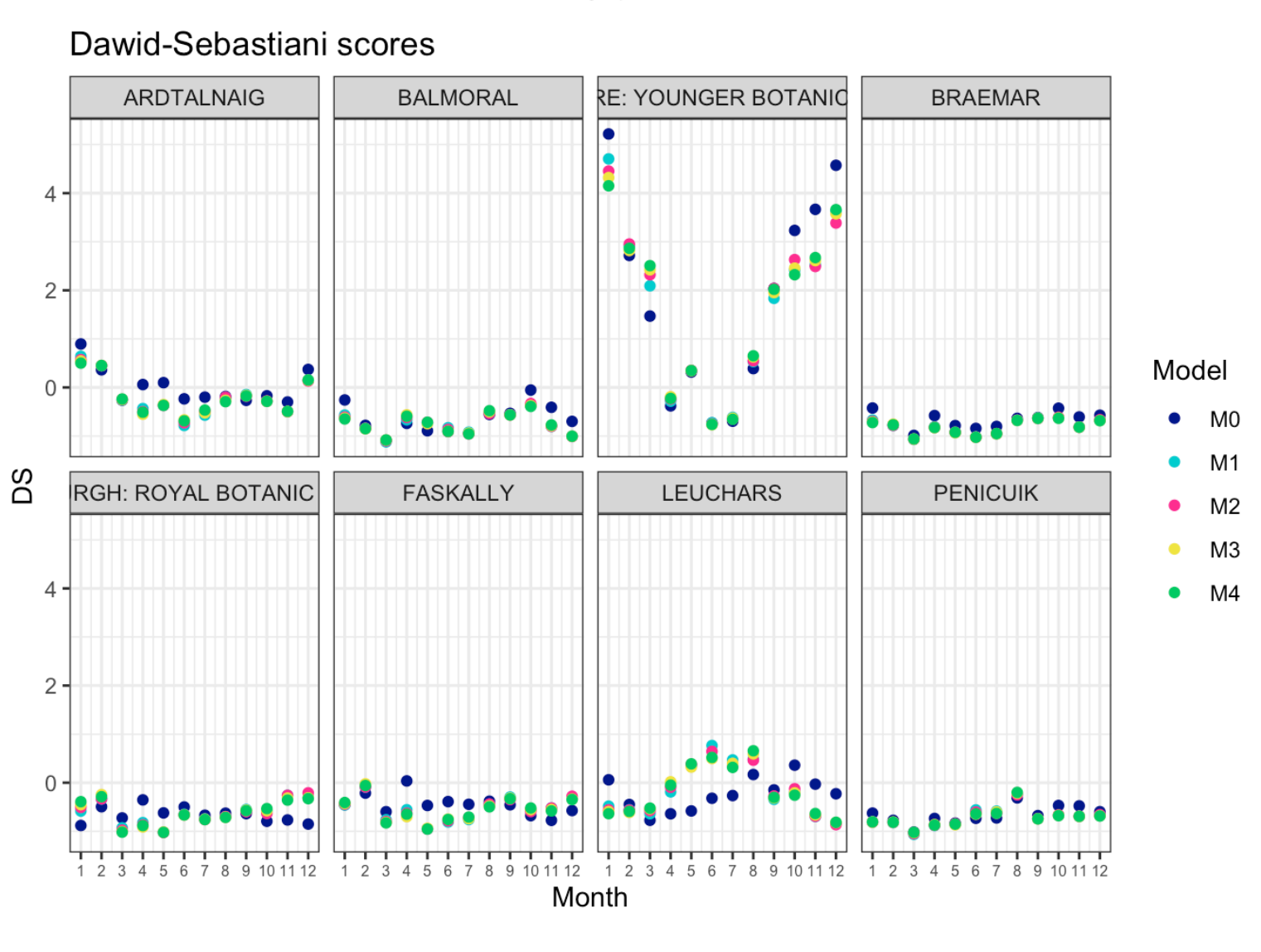
In the table, we can see that the value of the SE score and DS score for each station is quite similar across models M_1, M_2, M_3, and M_4, but for model M_0 they are quiet different. To view the difference more intuitively, we can create two subplots, one for the SE score and one for the DS score, to find out whether the models are equally effective at predicting the different stations.



The plot shows the same result as the table above, indicating that the value of both the SE score and DS score for all stations in the other four models are pretty similar, with slightly little fluctuation in the lines. However, the values for the M_0 model are quite different. These findings imply that compare to the other four models, the M_0 model lacks accuracy in predicting the precipitation at new location.

To further investigate whether the prediction accuracy is consistent throughout the year, we generate two plots that display the monthly average values of the two scores for each station of five models.





We found that the trend in both scores for each station is quite similar, and there exists visible gaps in the values of the two scores for each station between model M_0 and the other four models for each month. This suggests that the M_0 model lacks accuracy in predicting precipitation at new locations across the whole year. Additionally, it is obvious that the EDINBURGH: ROYAL BOTANIC GARDEN station is difficult to model, as the trend in both scores is significantly different from the trend of all other stations. Hence, the chance of encountering error increases when we apply stratified cross-validation that groups the data by station.

Code appendix

Quick-running code can instead be placed directly in the `.Rmd` file.

A	B
467.2	321.5946

Function definitions

```r
# Yichao Wang (s2056237)

# Place your function definitions that may be needed in the report.Rmd, including function documentation.
# You can also include any needed library() calls here
library(StatCompLab)
library(tidyverse)
library(ggplot2)
library(patchwork)
library(knitr)
library(dplyr)

###################

#' P-value and confidence interval
#' Construct the 95% confidence interval for each station
#'
#' The result will return a table includes the Name, p-value,
#' Monte Carlo standard deviation, and the CI for each station.
#'
#' @param data the desired data to be put into model
#' @param alpha the value of alpha, here we take 0.05 to construct a 95% CI
#' @param N the number of permutations
#'

p_value_CI <- function(data, alpha=0.05, N=10000) {
  # Initialize the count to be zero for each test statistic
  count <- rep(0, length = length(T_stats))
  for (n in 1:N) {
    # Generate a random permutation of the indices
    permutation <- sample(data$Summer, length(data$Summer), replace = FALSE)
    # Apply the permutation to the data and group by weather station
    ghcnd_permutation <- data %>%
      mutate("Summer" = permutation) %>%
      group_by(Name)
    # Compute the test statistic for the permuted data
    ghcnd_permutated <- ghcnd_permutation %>%
      summarise(Value = abs(mean(Value[Summer]) - mean(Value[!Summer])), .groups = "drop")
    # add one to the count if the test statistic is less than or equal to the observed value
    count <- count + (ghcnd_permutated$Value >= T_stats)
  }

  # Compute the p-value and Monte Carlo standard deviation
  p_value <- count / N
  monte_carlo_sd <- sqrt(p_value * (1 - p_value) / N)

  # Compute confidence interval for p-value
  lower_CI <- numeric(length(p_value))
  upper_CI <- numeric(length(p_value))
  for (i in 1:length(p_value)) {
    if (p_value[i] > 0) { # When x > 0 (p_value>0)
      lower_CI[i] <- p_value[i] - qnorm(1 - alpha/2) * monte_carlo_sd[i]
      upper_CI[i] <- p_value[i] + qnorm(1 - alpha/2) * monte_carlo_sd[i]
    } else { # When x = 0, CI=(0, 1-0.0025^(1/N))
      lower_CI[i] <- 0
      upper_CI[i] <- 1 - 0.025^(1/N)
    }
  }

  # Store the result in a dataframe and show them as a table
  result_table <- data.frame(Name = unique(winter_avg$Name),
                             p_value = p_value,
                             monte_carlo_sd = monte_carlo_sd,
                             lower_CI = lower_CI,
                             upper_CI = upper_CI)
  return(kable(result_table, align = "lcccc"))
}




#' Estimate models for the square root of the
#' monthly averaged precipitation values in Scotland
#'
#' The result will return the fitted model for five different
#' model M0, M1, M2, M3, M4 as different value of k entered
#'
#' @param data the desired data to be put into model
#' @param k the model M_k(k from 0 to 4)
#'

fit_model <- function(data, k) {
  # Define the formula for the model
  if(k == 0) { # When k = 0
    model <- Value_sqrt_avg ~ Longitude + Latitude + Elevation + DecYear
  } else{ # When k = 1,2,3,4
    model <- formula(paste0("Value_sqrt_avg ~ Longitude + Latitude + Elevation + DecYear + ",
                            paste0("cos(2*pi*", seq(1, k), "*DecYear) + sin(2*pi*", seq(1, k), "*DecYear)",
                                   collapse = " + ")))
  }
  # Fit the linear model
  fit <- lm(model, data = data)

  # Return the model object
  return(fit)
}




#' Stratified cross validation function
#'
#' The result will return a 96*5 dataframe shows the average SE and DS scores for each
#' month by station after the corss validation process, including the Name, ID, Month,
#' and its corresponding standard error scores(SE) and Dawid-Sebastiani scores(DS).
#'
#' since there are 8 stations and 12 months, we got a total of 12*8=96 rows.
#'
#' @param data the desired data to do the corss validation
#' @param k the model M_k(k from 0 to 4)

stratified_cv <- function(data, k) {
  pred2 <- data %>% mutate(mean = NA_real_, sd = NA_real_)
  for (id in unique(data$ID)) {
    if(k == 0) { # When k = 0
      model <- Value_sqrt_avg ~ Longitude + Latitude + Elevation + DecYear
    } else{ # When k = 1,2,3,4
      model <- formula(paste0("Value_sqrt_avg ~ Longitude + Latitude + Elevation + DecYear + ",
                              paste0("cos(2*pi*", seq(1, k), "*DecYear) + sin(2*pi*", seq(1, k), "*DecYear)",
                                     collapse = " + ")))
    }
    # fit and predict the model for the data
    fit <- lm(model, data %>% filter(ID != id))
    pred <- predict(fit, newdata = data %>% filter(ID == id), se.fit = TRUE)
    # evaluate the mean and sd
    pred2$mean[(pred2$ID == id)] <- pred$fit
    pred2$sd[(pred2$ID == id)] <- sqrt((pred$se.fit)^2 + (pred$residual.scale)^2)
    # mutate the table
    pred2 <- pred2 %>%
      group_by(ID, Name, Year, Month) %>%
      mutate(se = proper_score("se", Value_sqrt_avg, mean = mean),
             ds = proper_score("ds", Value_sqrt_avg, mean = mean, sd = sd))
  }
  # Calculate the average score for each month
  Average_score <- pred2 %>%
    group_by(ID, Name, Month) %>%
    summarise(se = mean(se), ds = mean(ds), .groups = "drop")
  return(Average_score)
}
```